

Day 2 Learning Summary: The SQL Language & Example Database

Introduction

SQL (Structured Query Language) is the standard language used to communicate with relational databases. It allows users to create databases, insert and modify data, retrieve information, manage permissions, and control transactions. Today focused on understanding SQL concepts, command categories, data types, and the example database that will be used throughout the learning journey.

1. What is SQL?

SQL (Structured Query Language) is a declarative language used to interact with relational databases.

Using SQL, you can:

- Create databases and tables
- Insert, update, and delete data
- Retrieve information
- Manage users and permissions
- Control database transactions

Key Points

- SQL tells the database **what** you want, not **how** to do it.
 - Most database systems (MySQL, PostgreSQL, SQL Server, Oracle, SQLite) use SQL with only minor syntax differences.
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2. Five SQL Command Families

Command Family	Full Form	Purpose	Common Commands
DDL	Data Definition Language	Defines database structure	CREATE, ALTER, DROP, TRUNCATE
DML	Data Manipulation Language	Adds or modifies data	INSERT, UPDATE, DELETE
DQL	Data Query Language	Retrieves data	SELECT

Command Family	Full Form	Purpose	Common Commands
DCL	Data Control Language	Manages user permissions	GRANT, REVOKE
TCL	Transaction Control Language	Manages transactions	COMMIT, ROLLBACK, SAVEPOINT

3. Common SQL Data Types

Numeric Types

Data Type	Purpose
INT	Whole numbers
DECIMAL(p,s)	Exact decimal values (recommended for money)
FLOAT / DOUBLE	Approximate decimal values

Tip: Use **DECIMAL** instead of **FLOAT** for financial data to avoid rounding errors.

Text Types

Data Type	Purpose
VARCHAR(n)	Variable-length text
CHAR(n)	Fixed-length text
TEXT	Long text

Date & Time Types

Data Type	Stores
DATE	Date only
TIME	Time only
DATETIME / TIMESTAMP	Date and time

Dates are commonly stored in the **YYYY-MM-DD** format.

Boolean Type

Stores logical values:

- TRUE
- FALSE

Some databases internally store them as:

- 1 = TRUE
 - 0 = FALSE
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NULL

`NULL` represents **missing or unknown data**.

Examples:

- Customer has no phone number.
- Product has no description.

Note: NULL is **not** the same as:

- 0
 - Empty string (`' '`)
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4. SQL Practice Environment

You can practice SQL using:

Online Platforms

- Browser-based SQL playgrounds (no installation required)

Local Installation

- SQLite + DB Browser
 - MySQL + MySQL Workbench
 - PostgreSQL + pgAdmin
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5. Example Database: online_store

The learning series uses a sample e-commerce database called **online_store**.

It contains five related tables:

Table	Purpose
customers	Customer information
categories	Product categories
products	Product details
orders	Customer orders
order_items	Products included in each order

6. Database Schema Overview

customers

Stores customer details such as:

- Customer ID
- Name
- Email
- City
- Country
- Signup Date

categories

Stores product categories.

Examples:

- Electronics
 - Books
 - Clothing
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products

Stores product information:

- Product ID
 - Product Name
 - Category
 - Price
 - Stock Quantity
-

orders

Stores order details:

- Order ID
 - Customer ID
 - Order Date
 - Order Status
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order_items

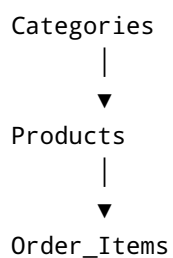
Stores products purchased in each order:

- Order Item ID
 - Order ID
 - Product ID
 - Quantity
 - Unit Price
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7. Table Relationships

The tables are connected using **Primary Keys (PK)** and **Foreign Keys (FK)**.

Relationship flow:





Relationships

- One category can have many products.
- One customer can place many orders.
- One order can contain many products.
- One product can appear in many orders.

The **order_items** table acts as a **bridge (junction) table**, connecting **orders** and **products** to handle many-to-many relationships.

8. Why Store Unit Price in order_items?

Although the **products** table contains the current product price, the **order_items** table stores the **price at the time of purchase**.

This preserves historical accuracy if product prices change later.

Key Takeaways

- SQL is the standard language for relational databases.
- SQL commands are grouped into **DDL, DML, DQL, DCL, and TCL**.
- Every column has a suitable data type.
- Use **DECIMAL** for monetary values.
- **NULL** represents missing data.
- Practice SQL using online playgrounds or local database software.
- The **online_store** database consists of five connected tables.
- Primary Keys and Foreign Keys establish relationships between tables.
- The **order_items** table resolves the many-to-many relationship between orders and products.

Day 2 Quick Recap

- ✓ What is SQL?
- ✓ SQL Command Families (DDL, DML, DQL, DCL, TCL)

- ✓ Common SQL Data Types
 - ✓ Understanding NULL
 - ✓ SQL Practice Environments
 - ✓ online_store Database Overview
 - ✓ Five Core Tables
 - ✓ Primary Key & Foreign Key Relationships
 - ✓ Junction (Bridge) Table Concept
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Interview Questions

1. What is SQL?
2. Why is SQL called a declarative language?
3. Explain the five SQL command families.
4. What is the difference between DDL and DML?
5. What is the purpose of DQL?
6. Why should DECIMAL be used for money instead of FLOAT?
7. What is NULL in SQL?
8. What are Primary Keys and Foreign Keys?
9. What is a junction (bridge) table?
10. Explain the relationships in the **online_store** database.